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Thermodynamic assessment of the Co–Ho binary system

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ABSTRACT

Based on the available experimental data of phase equilibria and thermodynamic properties from the literatures, the Co–Ho binary system has been thermodynamically assessed using the CALPHAD method. The solution phases, Liquid, FCC_A1, BCC_A2 and HCP_A3, were modeled as substitutional solution phases, of which the excess Gibbs energies were formulated with Redlich–Kister polynomial. Meanwhile, all intermetallic compounds were described as stoichiometric compounds. Subsequently, a set of self-consistent thermodynamic parameters describing various phases in this binary system has been obtained. The calculated results can reproduce well the corresponding experimental data.

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1. Introduction

Co-based alloys have attracted extensive attention owing to excellent magnetic properties [1]. As important alloying elements, rare earth elements (RE = Sm, Tb, Dy, Ho, Er, Tm, etc.) are often added into Co-based alloys to further improve their magnetic properties, especially magnetic coupling energy [2,3]. However, the mechanism with which rare earth elements can improve on performance of the Co-based alloys has not been clearly identified. Knowledge of phase diagram and thermodynamic properties of relevant Co–RE systems are particularly helpful to understand the role of rare earth element in the alloys and thus do help to the development of new Co-based alloys with superior performance.

To the best of our knowledge, no thermodynamic modeling has been reported for the Co–Ho binary system. The purpose of the present work is (1) to evaluate the measured phase diagram and thermodynamic data and (2) to provide a set of self-consistent parameters for calculation of the phase equilibria and thermodynamic properties in the Co–Ho binary system using the CALPHAD (CALculation of PHAse Diagram) method [4] and the Thermo-calc software package [5].

2. Evaluation of experiment information

Buschow and van der Goot [6] investigated the phase diagram of the Co–Ho binary system over the entire concentration range by means of X-ray diffraction, thermal analysis and metallography. Eight intermetallic compounds, namely $CoHo_3$, $CoHo_x$ ($x \approx 7/4$), Co_3Ho_4, Co_2Ho, Co_3Ho, Co_7Ho_2, $Co_{5.5}Ho$ and $Co_{17}Ho_2$, were identified in their work. Among these compounds, $CoHo_x$ ($x \approx 7/4$), Co_3Ho and $Co_{17}Ho_2$ melt congruently, while others formed by peritectic, eutectic or eutectoid reactions respectively. $Co_{5.5}Ho$ is unstable at low temperature and decomposes into Co_7Ho_2 and $Co_{17}Ho_2$ at about 1150 °C. A certain homogeneity region at elevated temperatures was found only for intermetallic compounds $Co_{5.5}Ho$ and $Co_{17}Ho_2$. However, the homogeneity regions for these two compounds were ignored in the present work due to the absence of experimental data. In addition, according to the crystal structure data of Schweizer [7], the compound, $CoHo_x$ ($x \approx 7/4$), presented by Buschow and van der Goot [6] was modified to Co_7Ho_{12}.

The solubility of Ho in FCC_A1(Co) or HCP_A3(Co) and that of Co in BCC_A2(Ho) or HCP_A3(Ho) were not reported in Ref. [6]. No other literatures have reported the measurement of solubility of terminal solid solutions. Thus, in the present optimization the solubility of each element in the Co–Ho terminal solid solutions was not taken into account.

Nikolaenko et al. [8] measured mixing enthalpies of liquid Co–Ho alloys at 1823 K by calorimetry. Using electromotive force (EMF) method, Su et al. [9] determined the Gibbs free energies of formation of intermetallic compounds $CoHo_3$, Co_2Ho, Co_3Ho,

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Table 1
T_c (K) and β (μ_B) for intermetallic compounds in the Co–Ho binary system.[a].

Magnetic parameters	$CoHo_3$[b]	Co_3Ho_4	Co_2Ho	Co_3Ho	Co_7Ho_2	$Co_{5.5}Ho$	$Co_{17}Ho_2$
T_c (K)	24	44	86.8	418	647	1033	1177
β (μ_B)	5.7	4.23	2.56	1.36	0.67	0.23	0.35

[a] The values of β (μ_B) are given in per mole of atoms.
[b] $CoHo_3$ is antiferromagnetic and its T_N and β values need to be multiplied by −3 when put into the PARROT program.

Co_7Ho_2 and $Co_{17}Ho_2$ at 1081 K. These experimental results [8,9] were adopted in the present optimization.

In the Co–Ho binary system, the intermetallic compounds Co_3Ho_4 [10], Co_2Ho [11–15], Co_3Ho [15–17], Co_7Ho_2 [15], $Co_{5.5}Ho$ [18–20] and $Co_{17}Ho_2$ [15,21] are ferromagnetic while the compound $CoHo_3$ [22,23] is antiferromagnetic. In order to model the magnetic contribution to the Gibbs energies, the data of the critical magnetic ordering temperature and the Bohr magnetic moment for intermetallic compounds are indispensable. Fortunately, Buschow [2] has compiled such data for the intermetallic compounds in this binary system (see Table 1) and they were used directly in the present optimization.

3. Thermodynamic model

3.1. Pure elements

The stable forms of the pure elements at 298.15 K and 1 bar were chosen as the reference states of the system. For the thermodynamic functions of the pure elements in their stable and metastable states, the phase stability for the element i in ϕ status is given as

$$^0G_i^{\phi}(T) = G_i^{\phi}(T) - H_i^{SER} = a + b\cdot T + c\cdot T\ln\ T + d\cdot T^2 + e\cdot T^3 + f\cdot T^{-1} + g\cdot T^7 + h\cdot T^{-9} \quad (1)$$

where H_i^{SER} is the molar enthalpy of the so-called "standard element reference" (SER), i.e., the enthalpies of the pure elements in their defined reference state at 298.15 K and 1 bar; T is the absolute temperature; $G_i^{\phi}(T)$ is the absolute molar enthalpy of the element i with structure ϕ. In thermodynamic study, absolute energy is not of importance, so the relative value of Gibbs energy $^0G_i^{\phi}(T)$ is adopted in CALPHAD approach. In the present work, the Gibbs energy functions of the pure elements Co and Ho, $^0G_{Co}^{\phi}(T)$ and $^0G_{Ho}^{\phi}(T)$ are taken from the SGTE (Scientific Group Thermodata Europe) database compiled by Dinsdale [24].

3.2. Solution phases

The substitutional solution model was employed to describe the solution phases including Liquid, FCC_A1, BCC_A2 and HCP_A3. The molar Gibbs energy of the solution phase ϕ (ϕ = Liquid, FCC_A1, BCC_A2 and HCP_A3) can be expressed as

$$G_m^{\Phi} = \sum x_i\,^0G_i^{\Phi} + RT\sum x_i\ \ln(x_i) + {}^EG_m^{\Phi} + {}^{mg}G_m^{\phi} \quad (2)$$

where G_m^{Φ} is the molar Gibbs energy of a solution phase ϕ; $^0G_i^{\Phi}$ is the molar Gibbs energy of the element i (i = Co, Ho) with the structure ϕ in a non-magnetic state; x_i the mole fraction of component i, R gas constant, T temperature; $^EG_m^{\Phi}$ the excess Gibbs energy, and $^{mg}G_m^{\Phi}$ is the magnetic contribution to the Gibbs energy which will be discussed in detail in Section 3.4. The excess Gibbs energy of phase ϕ can be expressed by the Redlich–Kister polynomial as [25]

$$^EG_m^{\Phi} = x_{Co}x_{Ho}\sum_{j=0}^{n}{}^{(j)}L_{Co,Ho}^{\Phi}(x_{Co} - x_{Ho})^j \quad (3)$$

Here $^{(j)}L_{Co,Ho}^{\Phi}$ is the interaction parameter between elements Co and Ho and is formulated as temperature dependent:

$$^{(j)}L_{Co,Ho}^{\Phi} = A_j + B_jT \quad (4)$$

where A_j and B_j are model parameters to be optimized in the present work.

3.3. Intermetallic compounds

All intermetallic compounds, $CoHo_3$, Co_7Ho_{12}, Co_3Ho_4, Co_2Ho, Co_3Ho, Co_7Ho_2, $Co_{5.5}Ho$ and $Co_{17}Ho_2$, were treated as stoichiometric phases in the Co–Ho binary system because no experimental data was made available for the homogeneity range of each compound. Gibbs energy of the compound Co_mHo_n is given as

$$G^{Co_mHo_n} = \frac{m}{m+n}\,^0G_{Co}^{hcp} + \frac{n}{m+n}\,^0G_{Ho}^{hcp} + \Delta G_f^{Co_mHo_n} + {}^{mg}G_m^{Co_mHo_n} \quad (5)$$

where $^0G_{Co}^{hcp}$ and $^0G_{Ho}^{hcp}$ are the Gibbs energies of the respective pure element Co and Ho in the non-magnetic hcp structure; $\Delta G_f^{Co_mHo_n}$ is the Gibbs energy of formation. Because the heat capacity of the compound is unavailable, the Neumann–Kopp rule is introduced, i.e. $C_p(Co_mHo_n) = (m/(m+n))C_p(Co) + (n/(m+n))C_p(Ho)$. Therefore, $\Delta G_f^{Co_mHo_n}$ is formulated as

$$\Delta G_f^{Co_mHo_n} = a + bT \quad (6)$$

where a and b are parameters to be optimized and $^{mg}G_m^{Co_mHo_n}$ is the magnetic contribution to the Gibbs energy discussed in Section 3.4.

3.4. Magnetic contribution to the Gibbs energy

The magnetic contribution to the Gibbs energy $^{mg}G_m$ can be described as

$$^{mg}G_m = RT\ln(\beta+1)f(\tau) \quad (7)$$

where τ is defined as T/T_c with T_c being the critical temperature for magnetic ordering, i.e. the Curie temperature (T_c) for ferromagnetic ordering and the Néel temperature (T_N) for antiferromagnetic ordering; β is a quantity related to the total magnetic entropy and is equal to the Bohr magnetic moment per mole of atoms in most cases; $f(\tau)$ represents the polynomials given by Hillert and Jarl [26] based on the magnetic specific heat of iron, i.e.

$$f(\tau) = 1 - \frac{79\tau^{-1}/140p + 474/497\left(1/p - 1\right)\times\left(\tau^3/6 + \tau^9/135 + \tau^{15}/600\right)}{A} \quad (8)$$

for $\tau < 1$ and

$$f(\tau) = -\frac{\tau^5/10 + \tau^{-15}/315 + \tau^{-25}/1500}{A} \quad (9)$$

for $\tau > 1$, where

$$A = \frac{518}{1125} + \frac{11{,}692}{15{,}975}\left(\frac{1}{p} - 1\right) \quad (10)$$

The parameter, p, which can be thought of as the fraction of the magnetic enthalpy absorbed above the critical temperature,

depends on the structure. $p = 0.4$ for BCC_A2 structure and $p = 0.28$ for the others.

For solution phase ϕ (ϕ = Liquid, FCC_A1, BCC_A2, HCP_A3), T_c^ϕ and β are described by the following expressions:

$$T_c^\phi = x_{Co}{}^0T_{cCo}^\phi + x_{Ho}{}^0T_{cHo}^\phi + x_{Co}x_{Ho}L_{T_c}^\phi \quad (11)$$

$$\beta_\phi = x_{Co}{}^0\beta_{Co}^\phi + x_{Ho}{}^0\beta_{Ho}^\phi + x_{Co}x_{Ho}L_\beta^\phi \quad (12)$$

where $L_{T_c}^\phi$ and L_β^ϕ are the magnetic interaction parameters between elements Co and Ho. They are set to zero due to lack of experimental data and extremely low solubility.

For intermetallic compounds, $CoHo_3$, Co_3Ho_4, Co_2Ho, Co_3Ho, Co_7Ho_2, $Co_{5.5}Ho$ and $Co_{17}Ho_2$, T_c^ϕ and β^ϕ were taken directly from the measured critical temperature for magnetic ordering and the mean Bohr magnetic moment per mole of atoms.

4. Optimization of thermodynamic parameters

On the basis of lattice stabilities cited from Dinsdale [24], the optimization of the model parameters was conducted using the PARROT module [27] in the Thermo-calc software package developed by Sundman et al. [5]. This module works by minimizing the square sum of the differences between experimental data and calculated values. In the optimization procedure, each set of experimental data was given a certain weight. The weights were changed systematically during the optimization until most of experimental data was accounted for within the claimed uncertainty limits.

The parameters of liquid phase were optimized first because many experimental data such as mixing enthalpy of liquid alloys and related phase boundaries are readily available. This was followed by the modeling of the parameters for terminal solution phases and intermetallic compounds. The evaluation was conducted one by one using the step-by-step optimization procedure. Finally, all known phases were optimized simultaneously considering all of the selected experimental information.

The lattice stability of Ho was used to optimize the parameters of terminal solution phases. The lattice stability of Ho in FCC_A1 status is absent in the SGTE database published by Dinsdale [24]. Fortunately, Wang et al. [28] has calculated the lattice stability of Ho in FCC_A1 status. Their calculations were carried out using the density functional theory in the generalized gradient approximation with projector augmented wave pseudopotential as implemented in the VASP package and give:

$${}^0G_{Ho}^{FCC_A1} = {}^0G_{Ho}^{HCP_A3} + 1180 \quad (13)$$

In addition, in the present work solubility was neglected in the FCC_A1, BCC_A2 and HCP_A3 phases. This was realized by assigning a large positive interaction parameter, i.e.

$${}^0L_{Co:Ho}^{BCC_A2} = {}^0L_{Co:Ho}^{FCC_A1} = {}^0L_{Co:Ho}^{HCP_A3} = 10^5 \quad (14)$$

Thermodynamic parameters for all condensed phases in the Co–Ho binary system obtained eventually in the present work were summarized in Table 2. Table 3 listed all invariant reactions for this binary system. Reasonable agreement has been realized between the calculated results and experimental data.

5. Results and discussion

Phase diagram of the Co–Ho binary system was calculated as illustrated in Fig. 1, and comparison of the calculated boundaries with experimental data reported by Buschow and van der Goot [6] was shown in Fig. 2. Enlarged parts of the Co–Ho phase diagram with experimental data were further shown in Figs. 3 and 4 respectively. In combination with Table 3, reasonable agreement has been obtained between the calculated and experimental data for phase boundaries. Most of the calculated reactions are within experimental error (about 6 K). Note here the temperatures of two reactions, L ↔ $CoHo_3$ + Co_7Ho_{12} at 1055 K and L ↔ Co_7Ho_{12} at 1056 K, deviate about 12 K from the corresponding measured values, 1043 K and 1068 K respectively. Nevertheless, these calculated reactions are still accepted if the measurement error and deviation caused by the volatility of rare earth elements are taken into account. In addition, care and great attention were paid to these two reactions during the optimization. If in the case that the temperatures were reproduced with an uncertainty of 5 K, unreasonable thermodynamic parameters might have to introduce.

As shown in Fig. 5, enthalpies of mixing of liquid alloys at 1823 K were calculated in comparison with experiment data measured by Nikolaenko et al. [8]. The calculated values are satisfactorily consistent with the measured data in Ref. [8].

Gibbs energies of formation of intermetallic compounds at 1081 K were further calculated and compared with experimental data obtained by Su et al. [9] in Fig. 6. Good agreement was realized between the calculated and experimental results for intermetallic

Table 2
Thermodynamic parameters of the Co–Ho binary system.

Phase	Thermodynamic parameters
Liquid	${}^0L_{Co,Ho} = -89{,}505 + 14.838 \times T$
	${}^1L_{Co,Ho} = 3171 - 14.4195 \times T$
FCC_A1	${}^0L_{Co,Ho}^{FCC_A1} = 10^5$
BCC_A2	${}^0L_{Co,Ho}^{BCC_A2} = 10^5$
HCP_A3	${}^0L_{Co,Ho}^{HCP_A3} = 10^5$
$CoHo_3$	$G^{CoHo_3} = 0.25^\circ G_{Co}^{Hcp} + 0.75^\circ G_{Ho}^{Hcp} - 12{,}669 + 0.32 \times T$
Co_7Ho_{12}	$G^{Co_7Ho_{12}} = 0.3684^\circ G_{Co}^{Hcp} + 0.6316^\circ G_{Ho}^{Hcp} - 16{,}400 + 0.4783 \times T$
Co_3Ho_4	$G^{Co_3Ho_4} = 0.4286^\circ G_{Co}^{Hcp} + 0.5714^\circ G_{Ho}^{Hcp} - 18{,}329 + 0.7603 \times T$
Co_2Ho	$G^{Co_2Ho} = 0.6667^\circ G_{Co}^{Hcp} + 0.3333^\circ G_{Ho}^{Hcp} - 24{,}705 + 0.6561 \times T$
Co_3Ho	$G^{Co_3Ho} = 0.75^\circ G_{Co}^{Hcp} + 0.25^\circ G_{Ho}^{Hcp} - 24{,}500 + 1.7643 \times T$
Co_7Ho_2	$G^{Co_7Ho_2} = 0.7778^\circ G_{Co}^{Hcp} + 0.2222^\circ G_{Ho}^{Hcp} - 23{,}958 + 2.4208 \times T$
$Co_{5.5}Ho$	$G^{Co_{5.5}Ho} = 0.8462^\circ G_{Co}^{Hcp} + 0.1538^\circ G_{Ho}^{Hcp} - 19{,}649 + 2.5519 \times T$
$Co_{17}Ho_2$	$G^{Co_{17}Ho_2} = 0.8947^\circ G_{Co}^{Hcp} + 0.1053^\circ G_{Ho}^{Hcp} - 18{,}532 + 4.026 \times T$

Note: Gibbs energies are expressed in J/mol. The lattice stability of Ho in FCC_A1 status is +1180 J versus HCP_A3 status by Wang et al. [28], while other lattice stabilities were given by Dinsdale [24].

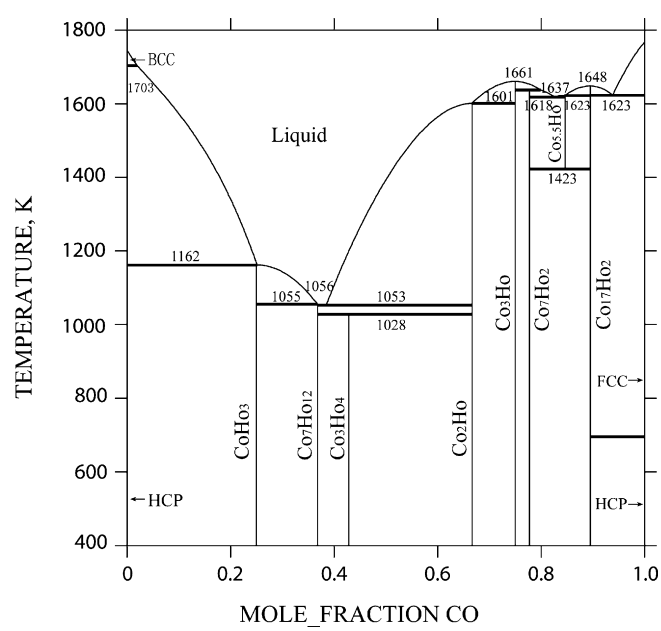

Fig. 1. Calculated phase diagram of Co–Ho binary system in this work.

Table 3
Invariant reactions in the Co–Ho binary system.

Reaction	Reaction type	T (K)	Composition of liquid (at.% Co)	Reference
L + BCC_A1 ↔ HCP_A3	Peritectic	1703	2.00	This work
L + HCP_A3 ↔ $CoHo_3$	Peritectic	1168	–	[6]
		1162	25.13	This work
L ↔ $CoHo_3$ + Co_7Ho_{12}	Eutectic	1043	33.00	[6]
		1055	36.78	This work
L ↔ Co_7Ho_{12}	Congruent	1068	–	[6]
		1056	36.84	This work
L ↔ Co_7Ho_{12} + Co_2Ho	Eutectic	1048	40.00	[6]
		1053	38.50	This work
Co_7Ho_{12} + Co_2Ho ↔ Co_3Ho_4	Peritectoid	1028	–	[6]
		1028	–	This work
L + Co_3Ho ↔ Co_2Ho	Peritectic	1603	–	[6]
		1601	66.57	This work
L ↔ Co_3Ho	Congruent	1663	–	[6]
		1661	75.00	This work
L + Co_3Ho ↔ Co_2Ho_7	Peritectic	1638	–	[6]
		1637	79.92	This work
L ↔ Co_7Ho_2 + $Co_{5.5}Ho$	Eutectic	1613	81.00	[6]
		1618	82.62	This work
L + $Co_{17}Ho_2$ ↔ $Co_{5.5}Ho$	Peritectic	1628	–	[6]
		1623	84.52	This work
$Co_{5.5}Ho$ ↔ Co_7Ho_2 + $Co_{17}Ho_2$	Eutectoid	1423	–	[6]
		1423	–	This work
L ↔ $Co_{17}Ho_2$	Congruent	1643	–	[6]
		1648	89.47	This work
L ↔ $Co_{17}Ho_2$ + FCC_A1	Eutectic	1618	93.00	[6]
		1623	93.80	This work

compounds Co_2Ho, Co_3Ho, Co_7Ho_2 and $Co_{17}Ho_2$. However, Gibbs energy of formation of $CoHo_3$ at 1081 K obtained through EMF measurements is $-31{,}275\,J\,mol^{-1}$, while the assessed value is $-11{,}953\,J\,mol^{-1}$, the difference is noticeable.

In order to judge the reliability of present calculated Gibbs energy of formation of $CoHo_3$, Gibbs energies of formation in other Co–RE (RE = rare earth elements) systems were adopted for comparison.

Enthalpies of formation of $CoDy_3$ and $CoGd_3$ at 298 K were respectively measured [29,30] to be $-8500\,J\,mol^{-1}$ and

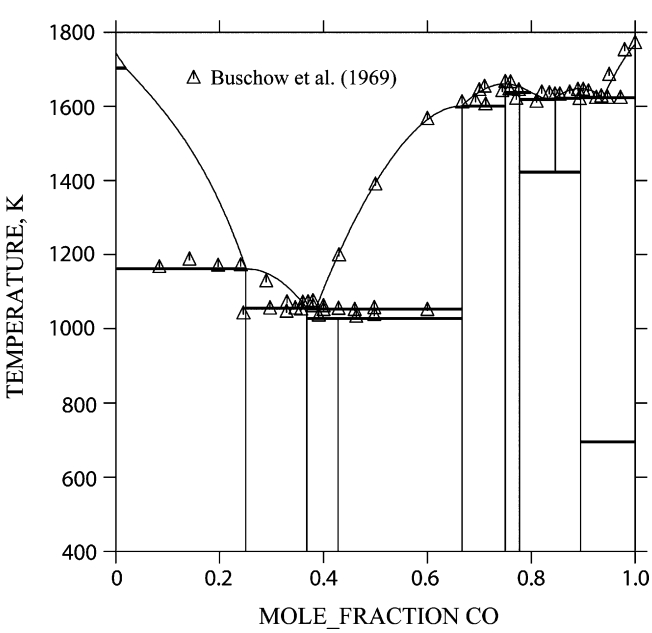

Fig. 2. Calculated phase diagram of Co–Ho binary system comparing with experimental data [6].

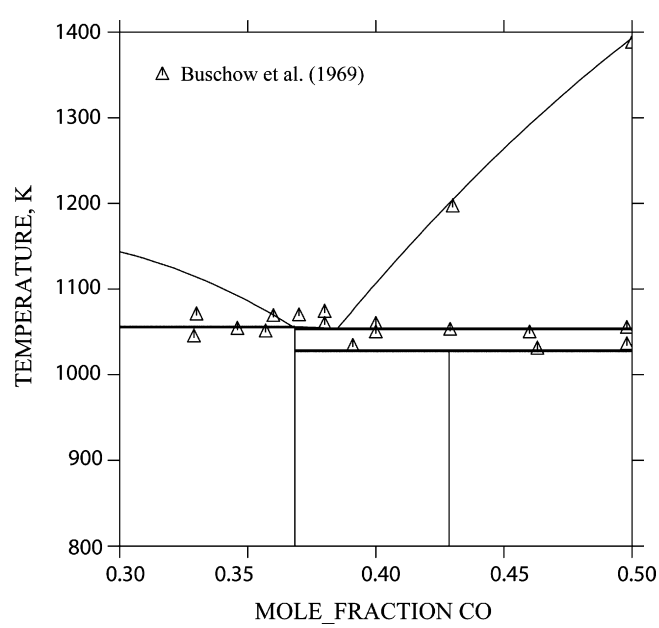

Fig. 3. Enlarged phase diagram of the Ho-rich part with experimental data [6].

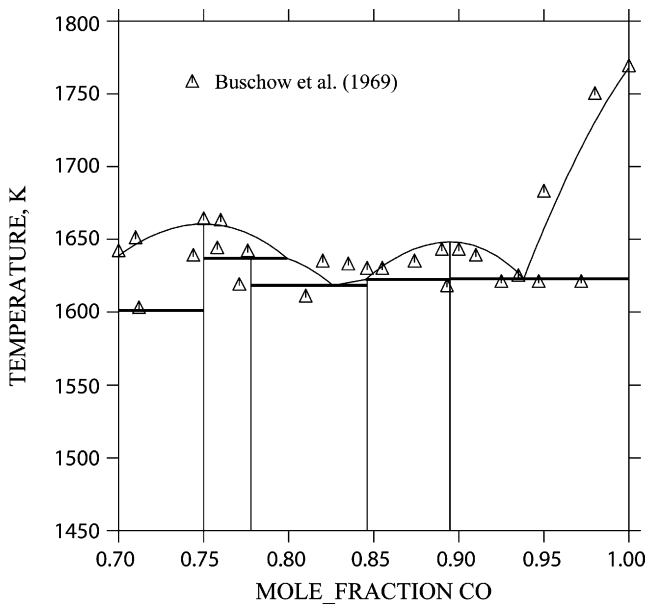

Fig. 4. Enlarged phase diagram of the Co-rich part with experimental data [6].

−8800 J mol^{-1}. Meanwhile, the enthalpies of formation of $CoRE_3$ (RE = Sm, Nd, Pr) at 298 K was calculated to be −10,000 J mol^{-1} in the Co–RE (RE = Sm, Nd, Pr) binary systems by Refs. [31–33]. Considering the similar chemical property between Ho and these RE, it can be concluded that the enthalpies of formation of $CoHo_3$ at 298 K should be around −10,000 J mol^{-1}, which is compatible with the presently assessed value, −10,539 J mol^{-1}. As shown in Fig. 7, enthalpies of formation of intermetallic compounds at 298 K in the Co–Ho binary system have been calculated. The enthalpy of formation of intermetallic compound Co_3Ho was the most negative, which is in accordance with the phase diagram that Co_3Ho has the highest melting point comparing with other compounds.

On the other hand, to satisfy both enthalpy of formation calculated in this work and Gibbs energy of formation measured by Su et al. [9] for intermetallic compound $CoHo_3$, its entropy of formation must become largely negative, which is not reasonable from thermodynamic viewpoint. Thus, the Gibbs energy of formation of intermetallic compound $CoHo_3$ measured by Su et al. [9] can be considered as not reliable. In summary, Gibbs energies of intermetallic compounds in the Co–Ho binary system have been reproduced reasonably well.

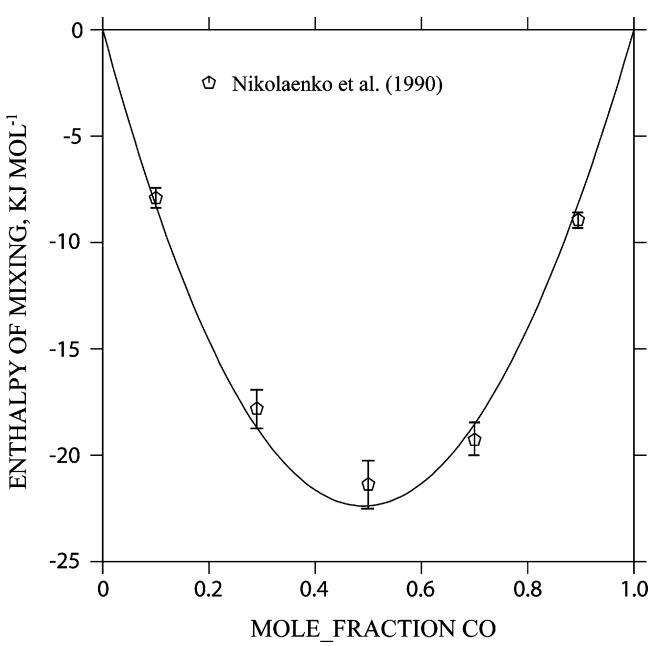

Fig. 5. Calculated enthalpies of mixing of liquid alloys at 1823 K compared with experimental data [8]. (Ref. states: liquid Co and liquid Ho.)

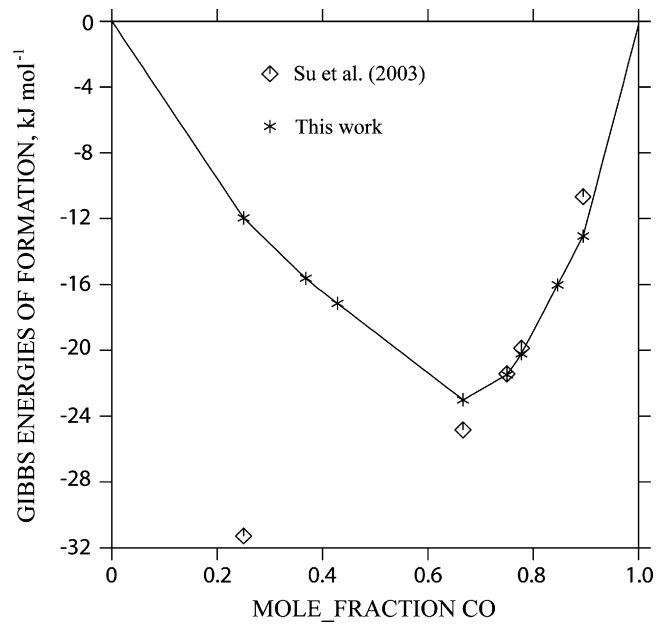

Fig. 6. Calculated Gibbs energies of formation of intermetallic compounds at 1081 K compared with experimental data [9].

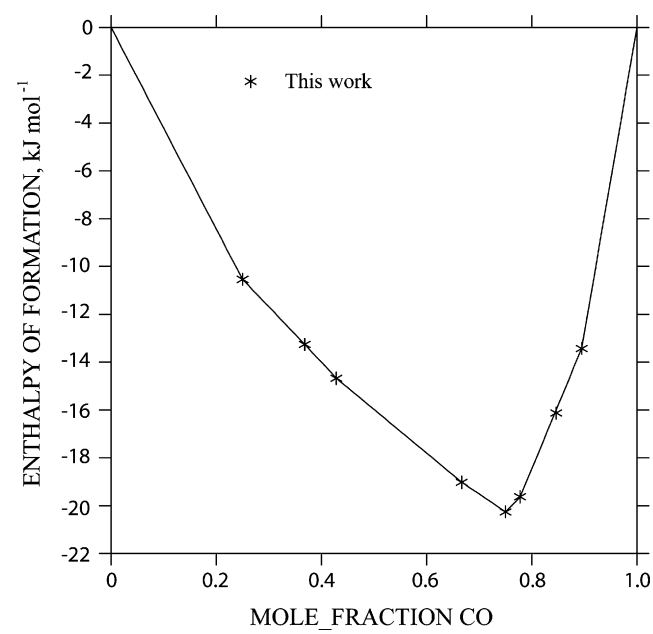

Fig. 7. Calculated enthalpies of formation of intermetallic compounds at 298 K.

6. Conclusions

The Co–Ho binary system has been thermodynamically optimized on the basis of available experimental information using the CALPHAD method. A set of self-consistent thermodynamic parameters has been obtained, which can reproduce most of the

experimental data of thermodynamic properties and phase diagram.

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